

ERIC MASTRO

Senior Engineering Lead, Technical & Strategic Direction

Greater Sydney Area, NSW, Australia · Open to remote

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🌐 egonat.me 🐙 github.com/emizzle 📺 in/ericmastro

SUMMARY

Senior engineering lead with over fifteen years' experience setting technical and strategic direction. The last eight years saw me building P2P networks, tooling, and an on-chain marketplace, incentivising honest behaviour through adversarial mechanism and protocol design. I worked with the executive board to define cross-org requirements, delivering against community expectations while leading the engineering team in a fully remote, open-source, and international company. Along the way, I presented and taught workshops at international conferences, translating highly technical concepts into digestible material for both technical and non-technical attendees. See my work at github.com/emizzle, with curated examples available at egonat.me.

KEY ACHIEVEMENTS

- **Led the marketplace team through adversarial mechanism and protocol design to handle real funds, with the company's reputation and its need to avoid legal ownership of those funds both on the line.** Planned, designed and implemented the marketplace client and EVM contracts while leading the team. The contracts were designed to incentivise honest behaviour while minimising exploit attack surface. Iterative research and design refinement culminated in a protocol that could durably store datasets in a permissionless network. Ultimately, this resulted in a unique design for token custody, co-authored as ERC-8233 (Draft), a proposed Ethereum standard for bounding what a compromised contract can do with escrowed funds.
- **Acted as interim Logos Storage lead through the cross-org transformation to operationally and technically align with the executive board's Logos vision.** Chaired working groups to define the roadmap and delivered specifications for the executive board. Improved speed to market by refactoring the codebase to a decentralised and private file-sharing client, while simultaneously driving the protocol's strategic pivot from an on-chain marketplace model to a gossip-based altruistic replication network.
- **Established a team work process improving output quality and effort accuracy, while creating end-to-end transparency for stakeholders.** Design write-ups improved the overall quality of output from the team by creating an artefact that could be iterated through open discussion before a single line of code was written, reducing wasted effort and technical debt. The design documents served as a base for planning sprints, as technical documentation, and the foundation for a specification once the work was complete. Once implemented, this process improved overall transparency for the executive board.
- **Improved project visibility across eight years of international conference talks and workshops,** bringing highly technical concepts within reach of non-technical participants, increasing the community size (an executive board goal). Since 2018, ten talks and workshops helped market the project, creating promotional material, broadening the project's reach and visibility, and showcasing thought leadership in the Ethereum world.
- **Led the first cross-org integration of Logos Storage** into Status Desktop for the community history archive service. Negotiated timelines with the executive board and worked across teams in the organisation to define and meet requirements for storage in their projects. Shipped C/Go bindings for Logos Storage, opening integration up to the entire community, meeting the executive board's expectations.
- **Since 2018: 796 pull requests authored · 934 reviewed · 1,639 commits across 95 repositories.**

EXPERIENCE

Logos (Institute of Free Technology / Status), Remote

April 2018 to September 2026 · 8 years 5 months, five progressive roles

Altruistic startup building fully open-source decentralised infrastructure. All work referenced is publicly available.

Engineering Lead, Logos Storage · Nov 2025 to Sep 2026

2 direct reports, 3 indirect · 202 commits · 95 PRs · 75 reviewed · Nim, JavaScript, C, C++, Shell, C#

- Drove the protocol's strategic pivot from an on-chain marketplace model to a gossip-based altruistic replication network. Chaired working groups to develop competing technical design proposals for the new direction.
- While acting as interim project lead, negotiated delivery timeframes, roadmap and requirements with executive leadership, successfully delivering FURPS, cross-org product integration, C/Go bindings, and specifications in less than four weeks.
- Ran cross-organisation requirements gathering with the Waku and Status Desktop teams, shaping FURPS requirements for those projects while writing our own.

- Led the integration of Logos Storage into Status Desktop for the community history archive service.
- Established a distributed testing framework pipeline in CI using an ephemeral K8s cluster.

Marketplace Team Lead, Codex • Jul 2024 to Nov 2025

4 direct reports • 197 commits • 118 PRs • 186 reviewed • Solidity, Nim, JavaScript

- Owned roadmap, FURPS, architecture and delivery for the on-chain storage market across the client, the smart contracts, documentation, specifications and L2 deployment.
- Ran iterative protocol R&D that resulted in an on-chain, permissionless durable storage network, allowing for real funds with legal implications.
- Co-developed the expanding-window mechanism that governs slot reservation eligibility using Kademlia distance.
- Redesigned state machine architecture after analysing recoverability during exceptions, cancellations and hardware failure. egonat.me/recoverability-analysis
- Drove the technical audit, including formal verification with Certora, and designed emergency preparedness procedures and contract monitoring in case of an exploit.
- Established a development process focussing on design iteration before writing code, improving output quality and stakeholder transparency, and serving as a foundation for documentation and specifications.
- Ran the Arbitrum L2 evaluation, modelling proof-verification economics against provider profitability.

Senior Distributed Networking Engineer, Codex • Oct 2021 to Jul 2024

314 commits • 147 PRs • 147 reviewed • Nim, Solidity, JavaScript

- Core contributor to the protocol node in Nim: peer discovery, content-addressed block storage, erasure coding, storage proofs and the marketplace client.
- Shipped slot reservations end-to-end, from research and design to implementation and sprint planning, deliberately holding three non-critical items to hit the deadline. egonat.me/how-i-work
- Co-authored nim-ethers (a Nim port of ethers.js, including EIP-1559 support for correct L2 gas estimation) and nim-serde, improving the serialisation story with a declarative pragma-driven API.
- Built nimv, a Nim toolchain manager fixing the failure mode that made the ecosystem standard painful: silently installing binaries incompatible with the host OS. Shipped via Homebrew, Chocolatey and apt.

Senior Software Engineer, Status Desktop • Apr 2020 to Oct 2021

188 commits • 176 PRs • 144 reviewed • Nim, C++, Qt (QML), JavaScript

- Built desktop features in Nim, C++, Qt and QML: community creation and editing, channel management, wallet transaction history, ENS validation, contract call encoding and a paid sticker market.
- Contributed to nim-status, a Nim rewrite of the status-go client library; worked on DOtherSide and nimqml, the Nim/C++ bindings letting a Qt application be driven from Nim; extended nim-sqlcipher, a Nim wrapper for encrypted SQLite, with data-access pragmas.

Senior Software Engineer, Embark Framework • Apr 2018 to Apr 2020

738 commits • 260 PRs • 382 reviewed • Node.js, JavaScript, TypeScript, Solidity

- Core contributor to Embark, an open-source Ethereum developer tooling framework.
- Introduced the RFC process the team adopted for design decisions, and diagnosed a code-generation race condition that had persisted over seven months.
- Built the npm plugin ecosystem covering MythX security analysis, Plasma chains, zkSNARKs and Status messaging, authored swarm-api for the Swarm storage platform, and wrote the workshop applications taught live at six international events: EDCON, CryptoLife, DappCon, ETHBerlin, Devcon and UNICEF Surge.

Earlier • Sydney and USA • 2007 to 2018

Senior .NET Full Stack Developer at Interesting (2010 to 2018) and Switch I.T. (2018); **.NET Developer** at Webcoda (2008 to 2010); **Software Engineer** at RedPrairie (2007 to 2008). C#, ASP.NET, JavaScript, TypeScript, SQL, C, VB.

- Technical lead on enterprise CMS builds for Sanitarium (Kentico Site of the Year 2016, Best Healthcare Site), Tourism NT, Lord Mayor's Charitable Foundation and Pickles Auctions. Introduced continuous delivery on Jenkins with Azure deployment slots, improving release reliability across the client portfolio.

EDUCATION

North Carolina State University. B.Sc. Computer Engineering; B.Sc. Electrical Engineering, *Magna Cum Laude*. Brazilian Jiu-Jitsu black belt (2017), Judo black belt (2007).